ISyE 6669 HW 14

1. Workers at the Execo manufacturing plant work 5 consecutive days a week, e.g. a worker can start work on a Monday and continue till Friday while another can start work on a Wednesday and continue till Sunday etc. Workers who are off on weekends (Saturday and Sunday) are paid \$300 per day (for each of their 5 work days), workers who's schedules involve one weekend day are paid \$320 per day (for each of their 5 work days), and workers who's schedules involve both weekend days are paid \$330 per day (for each of their 5 work days). The goal is to determine the number of workers for each schedule to minimize cost subject to the following restrictions:

   - There must be at least 25 workers each day.
   - There can be at most 35 workers on at least 3 of the 7 days of the week.
   - If the number of workers that start on a Monday exceed 10 then the number of workers available on a Wednesday should be no more than 28.
   - The number of workers that start on a Saturday should be less than the number of workers that start on a Monday or a Tuesday.
   - A total 40 workers are available.

   Formulate the above problem as a mixed-integer linear problem. Use x_i to denote the number (integer) of workers who start work on day i. You may need to introduce additional binary variables. If you use big-M numbers, please provide your best estimates of their values.

   Solution:

   We introduce an index i, where $i = 0$ represents Monday, $i = 1$ represents Tuesday, ..., and $i = 6$ represents Sunday. This indexing is used consistently throughout.

   Decision variables:

   - x_i: the number of workers who start work on day i (nonnegative integer variable)
   - y_i: the number of workers who are active on day i (nonnegative integer variable)

Objective function:

We minimize the total cost. Schedules without weekends (Monday start) pay \$300 per day, schedules with one weekend day (Tuesday or Sunday start) pay \$320 per day, and schedules with both weekend days (Wednesday, Thursday, Friday, or Saturday start) pay \$330 per day. Since each worker works for 5 consecutive days, the total cost per worker is the daily rate multiplied by 5. The objective function is

$$\min \quad 1500 \cdot x_0 + 1600 \cdot x_1 + 1650 \cdot x_2 + 1650 \cdot x_3 + 1650 \cdot x_4 + 1650 \cdot x_5 + 1600 \cdot x_6.$$

Nonnegativity and integrality:

$$x_i \geq 0, \quad y_i \geq 0, \quad x_i, y_i \in \mathbb{Z}, \quad \forall i = 0, 1, \ldots, 6.$$

Relationship between x_i and y_i:

Since each worker works for 5 consecutive days, the number of workers active on day i, denoted by y_i, is the sum of workers who are working on that day. Specifically:

$$\begin{aligned}
y_0 &= x_0 + x_3 + x_4 + x_5 + x_6, \\
y_1 &= x_0 + x_1 + x_4 + x_5 + x_6, \\
y_2 &= x_0 + x_1 + x_2 + x_5 + x_6, \\
y_3 &= x_0 + x_1 + x_2 + x_3 + x_6, \\
y_4 &= x_0 + x_1 + x_2 + x_3 + x_4, \\
y_5 &= x_1 + x_2 + x_3 + x_4 + x_5, \\
y_6 &= x_2 + x_3 + x_4 + x_5 + x_6.
\end{aligned}$$

Constraints:

(e) Total of 40 workers available:

$$x_0 + x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \leq 40.$$

(a) At least 25 workers each day: For all $i = 0, 1, \ldots, 6$,

$$y_i \geq 25.$$

(b) At most 35 workers on at least 3 days:

We introduce binary variables $p_i \in \{0, 1\}$ such that $p_i = 1$ when $y_i \leq 35$. This can be expressed by the following constraint:

$$y_i \geq 36(1 - p_i) + 25, \quad \forall i = 0, 1, \ldots, 6.$$

Since constraint (a) ensures $y_i \geq 25$, this representation is valid. When $p_i = 1$, we have $y_i \geq 25$, which is the same as constraint (a) and thus nonbinding. When $p_i = 0$, we have $y_i \geq 36$, which contradicts $y_i \leq 35$. Therefore, when $y_i \leq 35$, we must have $p_i = 1$.

Furthermore, the requirement that at least 3 days have at most 35 workers is expressed as

$$p_0 + p_1 + p_2 + p_3 + p_4 + p_5 + p_6 \geq 3.$$

(c) If $x_0 > 10$, then $y_2 \leq 28$:

We introduce a binary variable $q \in \{0, 1\}$ such that $q = 1$ when $x_0 > 10$. This can be expressed by the following constraint:

$$x_0 \leq 10 + 30q.$$

Since constraint (e) implies that the upper bound for x_i is 40, we use $30 \times q$ as the Big-M value. When $q = 0$, we have $x_0 \leq 10$, and when $q = 1$, we have $x_0 \leq 40$, which is nonbinding.

Next, we express the constraint that when $q = 1$, we must have $y_2 \leq 28$:

$$y_2 \leq 28 + 12(1 - q).$$

Since constraint (e) implies that y_i is also bounded above by 40, the right-hand side can be written as $28 + 12(1 - q)$ with the upper bound of 40. When $q = 1$, we have $y_2 \leq 28$, and when $q = 0$, we have $y_2 \leq 40$, which is nonbinding.

(d) $x_5 < x_0$ or $x_5 < x_1$:

In logical form, this is $x_5 < x_0$ or $x_5 < x_1$. We introduce auxiliary binary variables:

- $r_0 \in \{0, 1\}$: $r_0 = 1 \Leftrightarrow x_5 < x_0$
- $r_1 \in \{0, 1\}$: $r_1 = 1 \Leftrightarrow x_5 < x_1$

We need to model both directions of the equivalence.

For r_0:

$$\begin{aligned}
&x_5 < x_0 \Rightarrow r_0 = 1 \\
&x_0 - x_5 \leq 40r_0, \\
&\ r_0 = 1 \Rightarrow x_5 < x_0 \\
&x_5 - x_0 \leq -1 + 41(1 - r_0) = 40 - 41r_0.
\end{aligned}$$

For r_1:

$$\begin{aligned}
&x_5 < x_1 \Rightarrow r_1 = 1 \\
&x_1 - x_5 \leq 40r_1, \\
&\ r_1 = 1 \Rightarrow x_5 < x_1 \\
&x_5 - x_1 \leq -1 + 41(1 - r_1) = 40 - 41r_1.
\end{aligned}$$

The Big-M value 41 is chosen because $x_5 - x_0$ can be at most 40 (when $x_5 = 40$ and $x_0 = 0$), so $-1+41 = 40$ ensures the constraint is nonbinding when $r_0 = 0$.

Finally, we require that at least one condition is satisfied:

$$r_0 + r_1 \geq 1.$$

2. Jack and Jill love to watch movies and, as a couple, have vowed to always watch their movies together as a form of spending time with each other. They have subscribed for a 1 month free trial and estimated that they will watch 20 movies by the end of the free trial. They have pre-selected 100 movies that they would like to watch and now want to decide which 20 out of these 100 to watch during the free trial month. These 100 movies are divided as follows: Movies $1-20$ are action movies, Movies $21-40$ are romantic comedies, Movies $41-60$ are TV series, Movies $61-80$ are documentaries and Movies $81-100$ are science fiction.

   They associated an estimated common satisfaction index s_i to each movie and wish to maximize their estimated common satisfaction during this month. They also established the following constraints so that each of them will not be too unhappy: They must choose at least 3 action movies and at least 3 romantic comedies, but not more than 5 science fiction and no more than 2 documentaries.

   (assume that they will be able to watch all movies they request)

   a) Write an Integer programming formulation that Jack and Jill can use to solve this problem of maximizing their estimated common satisfaction. Clearly indicate your decision variables and what they mean.

      Solution:
      We introduce binary decision variables $\delta_i \in \{0,1\}$ for $i = 1, 2, \ldots, 100$, where $\delta_i = 1$ if movie i is watched and $\delta_i = 0$ otherwise.

      Objective function:
      We maximize the total satisfaction:

$$\max \quad \sum_{i=1}^{100} s_i \delta_i.$$

      Constraints:
      The total number of movies watched must be 20:

$$\sum_{i=1}^{100} \delta_i = 20.$$

      At least 3 action movies (movies 1–20) must be chosen:

$$\sum_{i=1}^{20} \delta_i \geq 3.$$

At least 3 romantic comedies (movies 21–40) must be chosen:

$$\sum_{i=21}^{40} \delta_i \geq 3.$$

At most 2 documentaries (movies 61–80) can be chosen:

$$\sum_{i=61}^{80} \delta_i \leq 2.$$

At most 5 science fiction movies (movies 81–100) can be chosen:

$$\sum_{i=81}^{100} \delta_i \leq 5.$$

b) Write one or more linear constraints to express the following additional conditions: (you may add decision variables if necessary, as long as you make it clear what they mean. Also, if you use a Big-M constant, please provide your best estimates of their values.)

i. Since movies 11, 12 and 13 are part of a series, either they get all of them, or they get none of them.

ii. Movies 41–50 are respectively seasons 1–10 of Friends, so they have decided that if they watch one season of Friends, they must watch all seasons before it. For example, if they watch season 6, they must also watch seasons 1–5.

iii. If they watch at least 2 movies out of 24, 34 and 77, then they must not watch more than one movie out of 27, 35 and 79.

Solution:

(i) Movies 11, 12, and 13 are part of a series:

Since movies 11, 12, and 13 are part of a series, either all of them are watched or none of them are watched. This is expressed as:

$$\delta_{11} = \delta_{12} = \delta_{13}.$$

(ii) Movies 41–50 are seasons 1–10 of Friends:

If they watch one season of Friends, they must watch all seasons before it. This means that if season k is watched (where $k \in \{41, 42, \ldots, 50\}$), then all seasons $41, 42, \ldots, k-1$ must also be watched. This is modeled by the following constraints:

$$\delta_{41} \geq \delta_{42} \geq \delta_{43} \geq \delta_{44} \geq \delta_{45} \geq \delta_{46} \geq \delta_{47} \geq \delta_{48} \geq \delta_{49} \geq \delta_{50}.$$

(iii) If at least 2 movies out of 24, 34, and 77 are watched, then at most one movie out of 27, 35, and 79 can be watched:

The logical statement is: if $\delta_{24}+\delta_{34}+\delta_{77} \geq 2$, then $\delta_{27}+\delta_{35}+\delta_{79} \leq 1$. We introduce an auxiliary binary variable $p \in \{0,1\}$ such that $p = 1$ when $\delta_{24} + \delta_{34} + \delta_{77} \geq 2$.
To express that $p = 1$ when $\delta_{24} + \delta_{34} + \delta_{77} \geq 2$, we use:

$$\delta_{24} + \delta_{34} + \delta_{77} \leq 1 + 2p.$$

When $\delta_{24} + \delta_{34} + \delta_{77} \geq 2$, we have $2 \leq 1 + 2p$, which forces $p = 1$. When $\delta_{24} + \delta_{34} + \delta_{77} \leq 1$, the constraint $1 \leq 1 + 2p$ allows p to be 0. Next, we express that when $p = 1$, we must have $\delta_{27} + \delta_{35} + \delta_{79} \leq 1$:

$$\delta_{27} + \delta_{35} + \delta_{79} \leq 1 + 2(1-p) = 3 - 2p.$$

When $p = 1$, this becomes $\delta_{27} + \delta_{35} + \delta_{79} \leq 1$, enforcing the "then" part. When $p = 0$, it becomes $\delta_{27}+\delta_{35}+\delta_{79} \leq 3$, which is nonbinding since each $\delta_i \leq 1$ and the sum is at most 3.

3. A power plant has four boilers. If a given boiler is operated, it can be used to produce a quantity of steam (in tons) between the minimum and maximum given in Table 1. The cost of producing a ton of steam on each boiler is also given, as well as the fixed cost of operating each boiler. Steam from the boilers is used to produce power on three turbines. If operated, each turbine can process an amount of steam (in tons) between the minimum and maximum given in Table 2. The cost of processing a ton of steam and the power produced by each turbine is also given. The power plant must produce at least 9,000 Kwh of power.

Boiler #	Min.	Max.	Cost per ton ($)	Fixed cost ($)
1	400	900	9	160
2	500	700	7	200
3	300	600	8	190
4	240	800	6	250

Table 1: Data for each of the boilers

Turbine #	Min.	Max.	Kwh per ton of steam	Processing Cost per Ton ($)
1	100	700	7	9
2	400	800	3	4
3	300	600	6	6

Table 2: Data for each of the turbines

Formulate a mixed integer linear program that can be used to minimize the cost while satisfying all constraints. Clearly specify what are the decision variables (and what they mean), as well as the objective function and constraints. Also, if you use a Big-M constant, please provide your best estimates of their values.

Solution:

Decision variables:

For boilers $i = 1, 2, 3, 4$:

- $u_i \in \{0, 1\}$: binary variable, equals 1 if boiler i is ON, 0 if OFF
- $x_i \geq 0$: continuous variable, the amount of steam (in tons) produced by boiler i
- $c_i \geq 0$: continuous variable, the cost of operating boiler i

For turbines $i = 1, 2, 3$:

- $v_i \in \{0, 1\}$: binary variable, equals 1 if turbine i is ON, 0 if OFF
- $y_i \geq 0$: continuous variable, the amount of steam (in tons) consumed by turbine i
- $z_i \geq 0$: continuous variable, the amount of power (in Kwh) produced by turbine i
- $p_i \geq 0$: continuous variable, the cost of processing steam on turbine i

Objective function:

We minimize the total cost:

$$\min \quad \sum_{i=1}^{4} c_i + \sum_{i=1}^{3} p_i.$$

Constraints:

Boiler constraints:

For each boiler $i = 1, 2, 3, 4$, the steam production x_i must be between the minimum and maximum values when the boiler is ON, and must be 0 when OFF:

$$\begin{aligned}
400u_1 \leq x_1 \leq 900u_1, \\
500u_2 \leq x_2 \leq 700u_2, \\
300u_3 \leq x_3 \leq 600u_3, \\
240u_4 \leq x_4 \leq 800u_4.
\end{aligned}$$

The cost for each boiler i consists of a fixed cost and a variable cost:

$$\begin{aligned}
c_1 &= 160u_1 + 9x_1, \\
c_2 &= 200u_2 + 7x_2, \\
c_3 &= 190u_3 + 8x_3, \\
c_4 &= 250u_4 + 6x_4.
\end{aligned}$$

Turbine constraints:

For each turbine $i = 1, 2, 3$, the steam consumption y_i must be between the minimum and maximum values when the turbine is ON, and must be 0 when OFF:

$$
\begin{aligned}
100v_1 \leq y_1 \leq 700v_1, \\
400v_2 \leq y_2 \leq 800v_2, \\
300v_3 \leq y_3 \leq 600v_3.
\end{aligned}
$$

The power produced by each turbine i is proportional to the steam consumed:

$$
\begin{aligned}
z_1 &= 7y_1, \\
z_2 &= 3y_2, \\
z_3 &= 6y_3.
\end{aligned}
$$

The processing cost for each turbine i is:

$$
\begin{aligned}
p_1 &= 9y_1, \\
p_2 &= 4y_2, \\
p_3 &= 6y_3.
\end{aligned}
$$

Steam balance constraint:

The total steam produced by all boilers must be at least the total steam consumed by all turbines:

$$\sum_{i=1}^{4} x_i \geq \sum_{i=1}^{3} y_i.$$

Power production constraint:

The total power produced must be at least 9,000 Kwh:

$$\sum_{i=1}^{3} z_i \geq 9000.$$

4. Consider the polytope P given by:

$$
\begin{aligned}
2x_1 + 3x_2 &\leq 100 \\
x_1 &\geq 0 \\
x_2 &\geq 0.
\end{aligned}
$$

Notice that the point $(5, 10)$ is a feasible point for P. Write an integer programming formulation whose feasible set is exactly the set of integer points in P except $(5, 10)$. (No need to give any objective function.)

Solution:

We introduce binary variables $\delta_1, \delta_2 \in \{0, 1\}$ such that:

$$x_1 = 5 \Rightarrow \delta_1 = 1, \quad x_2 = 10 \Rightarrow \delta_2 = 1.$$

To exclude $(5, 10)$, we require:

$$\delta_1 + \delta_2 \leq 1.$$

Modeling $x_1 = 5 \Rightarrow \delta_1 = 1$:

Taking the contrapositive: $\delta_1 = 0 \Rightarrow x_1 \neq 5$. Since $x_1 \in \mathbb{Z}_{\geq 0}$, we have:

$$\delta_1 = 0 \Rightarrow (x_1 \geq 6 \text{ XOR } x_1 \leq 4).$$

To model the XOR, we introduce auxiliary binary variables $p_1, p_2 \in \{0, 1\}$ such that:

$$p_1 = 1 \Leftrightarrow x_1 \geq 6, \quad p_2 = 1 \Leftrightarrow x_1 \leq 4.$$

For p_1:

$$\begin{aligned}
&x_1 \geq 6 \Rightarrow p_1 = 1 \quad \text{(contrapositive: } p_1 = 0 \Rightarrow x_1 \leq 5)\\
&\qquad x_1 \leq 5 + 45p_1,\\
&p_1 = 1 \Rightarrow x_1 \geq 6\\
&\qquad x_1 \geq 6p_1.
\end{aligned}$$

For p_2:

$$\begin{aligned}
&x_1 \leq 4 \Rightarrow p_2 = 1 \quad \text{(contrapositive: } p_2 = 0 \Rightarrow x_1 \geq 5)\\
&\qquad x_1 \geq 5(1 - p_2),\\
&p_2 = 1 \Rightarrow x_1 \leq 4\\
&\qquad x_1 \leq 50 - 46p_2.
\end{aligned}$$

The XOR condition is expressed as:

$$p_1 + p_2 = 1 - \delta_1.$$

The Big-M values are chosen as: $x_1 \leq 50$ (from $2x_1 + 3x_2 \leq 100$ with $x_2 = 0$), so $45 = 50 - 5$ and $46 = 50 - 4$.

Modeling $x_2 = 10 \Rightarrow \delta_2 = 1$:

Taking the contrapositive: $\delta_2 = 0 \Rightarrow x_2 \neq 10$. Since $x_2 \in \mathbb{Z}_{\geq 0}$, we have:

$$\delta_2 = 0 \Rightarrow (x_2 \geq 11 \text{ XOR } x_2 \leq 9).$$

To model the XOR, we introduce auxiliary binary variables $q_1, q_2 \in \{0, 1\}$ such that:

$$q_1 = 1 \Leftrightarrow x_2 \geq 11, \quad q_2 = 1 \Leftrightarrow x_2 \leq 9.$$

For q_1:

$$\begin{aligned}
x_2 \geq 11 \Rightarrow q_1 = 1 \quad &(\text{contrapositive: } q_1 = 0 \Rightarrow x_2 \leq 10) \\
x_2 \leq 10 + 23q_1, \\
q_1 = 1 \Rightarrow x_2 \geq 11 \\
x_2 \geq 11q_1.
\end{aligned}$$

For q_2:

$$\begin{aligned}
x_2 \leq 9 \Rightarrow q_2 = 1 \quad &(\text{contrapositive: } q_2 = 0 \Rightarrow x_2 \geq 10) \\
x_2 \geq 10(1 - q_2), \\
q_2 = 1 \Rightarrow x_2 \leq 9 \\
x_2 \leq 33 - 24q_2.
\end{aligned}$$

The XOR condition is expressed as:

$$q_1 + q_2 = 1 - \delta_2.$$

The Big-M values are chosen as: $x_2 \leq 33$ (from $2x_1 + 3x_2 \leq 100$ with $x_1 = 0$), so $23 = 33 - 10$ and $24 = 33 - 9$.